

Two Field Spread: numerical algorithms

v0.5.6 implementation and numerical assessment

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Companion paper. Christopher Angstmann, Derick Diana and Tim Gebbie, *A Finite Bid–Ask Spread from Replenishment Displaced from the Quote*. Forthcoming preprint.

Status. The Markov solver, execution tape and long-run diagnostic correlations are implemented and tested. Numerical convergence and scientific acceptance remain pending; fractional transport and calendar subordination are future work. This draft describes the actual software; it does not amend the supplied theory paper or letter.

1 Model, agents and the inherited numerical construction

The state is two nonnegative densities, $\rho_B(x_j, u_n)$ and $\rho_A(x_j, u_n)$, in one market. A cell carries volume $\rho_i \Delta x$; x is log price. Fixed endpoint reservoirs are excluded from executable volume. Lit and latent sources aggregate supply; diffusion aggregates repricing; cancellation removes standing volume; symmetric reaction removes overlap. These are an aggregate closure with an order-level random-walk interpretation. The current solver does not simulate individual traders, wealth, beliefs or strategy choice.

The market-making sector is represented by pending completion stocks and source placement. All explicitly executed child volume in this pilot is attributed to that sector. Its opposite-side source is *completion-equivalent*: survival and execution probabilities are assumed inside the completion/placement specification. Source delivery is neither a new trade print nor independently observed settlement. Reaction loss is accounted for as a field sink, without inventing aggressor signs or prices for it. Thus the trade tape records the specified aggressor programme, not every possible microscopic match.

Let Q_B denote bid replenishment owed after aggressive buys consuming the ask, and Q_A ask replenishment owed after aggressive sells consuming the bid. The quote-setting sample obeys

$$Q_\Sigma = Q_A + Q_B, \quad Q_\Delta = Q_A - Q_B, \quad (1)$$

$$\Sigma = \Sigma_0 + \chi_s Q_\Sigma, \quad \mu = \mu_0 - \chi_m Q_\Delta, \quad q_b = \mu - \Sigma/2, \quad q_a = \mu + \Sigma/2. \quad (2)$$

Observed quotes p_b, p_a are side-density threshold crossings, not q_b, q_a . Zero signed inventory does not imply zero total pending exposure. Also, $\Sigma_0 = 12$ is imposed in this pilot: the existence of its baseline spread is not evidence of spontaneous spread formation from zero standoff.

Diana–Gebbie explicitly identify the need for two densities to represent a nonzero spread [1]. The correlation-emergence software supplies the operational DTRW, execution-record and observation conventions [2, 3]. Its two coupled books are not identified with the two sides of this market. The new density core implements the supplied spread paper’s Markov branch. For the common reaction-free interior operator, the prototype has

$$\rho^{n+1} = e^{-\nu \Delta u} \rho^n + \frac{r}{2} \delta_x^2 \rho^n + \Delta u S^n, \quad r = 2D\Delta u / \Delta x^2,$$

where $\delta_x^2 \rho_j = \rho_{j-1} - 2\rho_j + \rho_{j+1}$. Here cancellation is $(1 - \nu \Delta u) \rho^n$. Their difference is $(e^{-\nu \Delta u} - 1 + \nu \Delta u) \rho^n$, of one-step order Δu^2 . Matched transport agrees to roundoff at $\nu = \kappa = 0$; complete trajectories are not asserted identical. The prototype consumes before its field update; the present paper convention consumes after it. Phase labels preserve this distinction. See `compatibility-v0.4.1.json` for the executed comparison.

2 Algorithm 1: implemented Markov update

Inputs: previous post-event densities, post-event pending stocks P_B, P_A , fixed $\Delta x, \Delta u$, parameters and event-integrated requests. Both sides read one frozen field; new executions cannot complete immediately.

1. Deliver old pending cohorts: $C_i = (1 - e^{-\Delta u/\tau_h})P_i$ and $Q_i = P_i - C_i$. For the next-update control set $C_i = P_i$. This is the only “immediate” convention used. Form q_b, q_a from Q above.
2. Evaluate the paper’s lit and latent sources S_i^{ext} on the outward side of q_i . Normalize uniform nodal weights g_B on $[q_b - w, q_b]$ and g_A on $[q_a, q_a + w]$ so that $\Delta x \sum_j g_{i,j} = 1$. Reject empty or reservoir-truncated supports. No supply is placed between the placement edges.
3. For interior nodes, compute the pre-event field

$$\tilde{\rho}_{i,j} = (1 - 2a - \nu\Delta u - \kappa\Delta u\rho_{i,j})\rho_{i,j} + a(\rho_{i,j-1} + \rho_{i,j+1}) + \Delta u S_{i,j}^{\text{ext}} + C_i g_{i,j}, \quad a = D\Delta u/\Delta x^2.$$

Require $2a + \nu\Delta u + \kappa\Delta u\rho_{i,j} \leq 1$ everywhere. Keep endpoint reservoirs fixed; reject inadmissible states without clipping.

4. Read pre-event quotes by linear interpolation at density threshold ϵ : the rightmost downward bid crossing and leftmost upward ask crossing. Record crossing counts; reject a selected plateau, insufficient selected slope, reservoir failure or contact.
5. Execute each request in outward price priority, starting at its pre-event quote and ending at the configured execution depth. Restrict to interior node centres. Remove at most $\tilde{\rho}_{i,j}\Delta x$ per node; record actual fills, requests and unfilled demand. No reservoir is consumed. The current tape requires one aggressor side per event.
6. Let v_A be actual buy volume and v_B actual sell volume. Set $P_B^{\text{new}} = Q_B + v_A$, $P_A^{\text{new}} = Q_A + v_B$. Read post-event quotes and save both inventory samples, actual removals and separate source increments. An event at u is applied at the end of the interval ending at u ; pre/post fields share u with different phase labels.

Checks on every update. For each side, the interior volume change is boundary flux minus cancellation, minus symmetric reaction, plus external supply, plus completion, minus execution. The boundary increment is $D\Delta u(\rho_0 - \rho_1 + \rho_J - \rho_{J-1})/\Delta x$. Reaction loss is $\kappa\Delta u\Delta x \sum_j \rho_{A,j}\rho_{B,j}$ on each side. It cancels from imbalance and occurs twice in total density. Cumulative initiation must equal cumulative completion plus post-event pending stock, separately by side.

Initialization and recovery. Relax the same positive-reaction equations with empty pending stock and no executions to the configured residual. Recompute geometric completion weights when Δu changes, retaining physical τ_h . Preserve pending tails beyond a finite plotting horizon. Volume-balanced round trips need not restore liquidity at the quote at the same time: delivery occurs elsewhere and transport takes time. Neither the ledger nor a stationary field establishes dealer profitability or a strategic equilibrium. Hard source supports require grid-phase refinement; the current late spread offset has not been accepted as physical.

3 Algorithm 2: implemented execution tape

1. For event n retain nonzero executed node volumes $v_{n,j}$, parent identifier, aggressor sign ε_n (+1 buy, -1 sell), operational index and time. No fill means no trade observation. Preserve each fill row and the requested/unfilled amounts in the event record.
2. Following the prior executable convention, compute the primary execution log price

$$z_n = \frac{\sum_j v_{n,j} x_j}{\sum_j v_{n,j}}.$$

Store e^{z_n} as the geometric execution price and $\sum_j v_{n,j} e^{x_j} / \sum_j v_{n,j}$ as the separate arithmetic price VWAP. Do not call these identical. Store pre/post bid, ask, log midpoint and spread, and the placement and pending state that generated the event.

3. Preserve the true aggressor sign separately from the quote sign $\text{sign}(z_n - m_n^{\text{pre}})$ and tick sign $\text{sign}(z_n - z_{n-1})$. The first tick label is zero; a zero tick subsequently carries the last nonzero tick label, as in v2.0.0.
4. Using consecutive executed events, calculate $r_n^m = m_n^{\text{post}} - m_{n-1}^{\text{post}}$, $r_n^T = z_n - z_{n-1}$ and $\Delta s_n = s_n^{\text{post}} - s_{n-1}^{\text{post}}$. Retain absolute increments. First increments are missing, not zero. An instantaneous pre/post midpoint jump is stored separately from r_n^m , which also includes inter-event field evolution.

Prices are measured from actual density consumption; there is no independent synthetic price process. The current six same-sign children demonstrate the measurement path only. Their sign variance is zero, so no sign ACF or LMF claim is available from this run.

4 Algorithm 3: lagged correlation

The estimator and the registered long-run protocol below are implemented. The resulting correlations remain diagnostic pending numerical acceptance. For aligned finite series X, Y , define

$$\hat{C}_{XY}(k) = \frac{\sum_{n \in I_k} (X_n - \bar{X}_k)(Y_{n+k} - \bar{Y}_k)}{\sqrt{\sum_{n \in I_k} (X_n - \bar{X}_k)^2 \sum_{n \in I_k} (Y_{n+k} - \bar{Y}_k)^2}}.$$

Each mean is taken over its own overlapping slice. Positive k means X leads Y . Save pair counts, lag units and phase alignment. Zero variance is undefined, not zero correlation. Do not silently compress missing observations. This reproduces the v2.0.0 executable Pearson ACF. Its supplementary text instead prints a global-mean covariance normalization; that documentation mismatch is not carried into this supplement.

The long runs retain raw price paths and use $r^m, r^T, |r^m|, |r^T|$ for price ACFs. For a positive price, $|P| = P$; an absolute log-price level depends on the arbitrary origin. Spread-level ACFs require a checked stationary window. Price-level ACFs, if retained, must be labelled descriptive rather than evidence of long memory. A fixed valuation centre and fixed reservoirs can cause mean reversion; diffusive prices are not guaranteed.

5 Algorithm 4: registered flow and market-maker controls

No chartist or fundamentalist population is required for the current spread question. Static sources do not already encode strategies; fixed μ_0 is a valuation reference. The retained closure is unchanged.

1. For the finite programme, execute six children of volume 0.05 at $u = 1, 1.4, 1.8, 2.2, 2.6, 3$, either all buys or alternating buys/sells. Compare moving placement, next-update completion, both feedbacks off, $\chi_s = 0$ and $\chi_m = 0$. Use identical actual absolute fills. Show total and signed pending stocks separately; delivery is not a second trade.
2. For long runs, draw integer parent length $L = \min(\lfloor 2(1 - U)^{-1/1.5} \rfloor, 512)$ with U uniform on $[0, 1)$. Draw independent centred parent signs. Sample the initial length from $\Pr(L_0 = l) = l \Pr(L = l) / E[L]$, with uniform integer age $0, \dots, l - 1$. Keep distinct parent IDs even when adjacent signs agree.
3. Execute children of volume 0.002 every 0.01 operational units. Use seeds 41001–41008, 2048 burn events and 8192 retained events per path. Compare moving placement with the same splitting tapes at fixed placement, and independent signs with seed offset 100000 and matched volume/rate. The first retained increment uses the preceding burn-event price.
4. Compute ACFs within each path for $\varepsilon, r^m, r^T, |r^m|, |r^T|, s$. Compute CCFs for (ε, r^m) , $(\varepsilon, \Delta s)$, (r^m, r^T) , $(r^m, \Delta s)$, $(|r^m|, s)$ and $(|r^T|, s)$ through lag 128. Average eight independent paths; bands are descriptive mean $\pm$ twice the between-path standard error. Paired controls are not extra replicates. Save pair counts, per-path correlations and spread half-window means.

The exact stationary single-active-parent sign reference is

$$C_\varepsilon(k) = \frac{E[(L - k)_+]}{E[L]}.$$

For an untruncated tail with $1 < \beta < 2$ this renewal benchmark has exponent $\beta - 1$, consistent with the order-splitting motivation [4]. The implemented capped law is not the full concurrent-parent LMF population, and cannot establish asymptotic long memory. The configuration reserves lags 8–64 for a possible fit; no exponent has been fitted or accepted. Persistent signs are declared input, not memory generated by density diffusion.

Grids and retained data. F2/F3 use $(\Delta x, \Delta u) = (0.025, 0.00025)$. F4 shows both/fixed/width-only balanced controls at $\Delta x = 0.003125$ and $\Delta u = 0.00000390625$. The 12-run assessment crosses $\Delta x = 0.00625, 0.003125$ at this common step, phases 0 (on nodes) and 0.5 (between nodes), and three feedback settings. Thirty-nine earlier finite controls remain saved. Centre-only and interaction comparisons remain at their previous resolution. Long-run groups and all 26 long paths are unchanged. Half-cell padding preserves reflection while moving each reservoir by $\Delta x/2$; separate domain checks are retained.

Figure inventory. F2: density snapshots; F3: prices and market-maker state; F4: matched mechanisms and flow; V1: held-state profile video. F5 has aligned execution-price, spread and sign paths plus the six ACFs. F6 has the six CCFs. Every panel uses retained simulation data or the declared sign-input reference. No independent theory gallery is required.

6 Targeted response refinement in v0.5.6

All 29 focused tests pass. Twelve balanced controls retain the six alternating 0.05 children, horizon 8 and completion law. Each grid initializes independently to rate-residual 10^{-8} . Settings are both feedbacks, fixed placement and width only ($\chi_m = 0$). At $\Delta x = 0.00625$, compare the new $\Delta u = 0.00000390625$ against retained $\Delta u = 0.000015625$ (time). Compare $\Delta x = 0.00625, 0.003125$ at the new common step (space).

For $y = m, s$, retain $R_k(u) = y_k(u) - y_k(0)$ and the matched effect $E_k(u) = R_k(u) - R_{\text{fixed}}(u)$. Report their grid differences without allocating additive causal shares. Matched fills give equal pending schedules; the evolving densities can differ.

Check	Phase	Feedback	Max. spread diff.	Max. response diff.
Time	0.0	Both	0.000015	0.000015
Space	0.0	Both	0.005198	0.004393
Time	0.0	Fixed	0.000015	0.000015
Space	0.0	Fixed	0.005198	0.002078
Time	0.0	Width only	0.000015	0.000015
Space	0.0	Width only	0.005198	0.004392
Time	0.5	Both	0.000015	0.000015
Space	0.5	Both	0.002383	0.002378
Time	0.5	Fixed	0.000015	0.000015
Space	0.5	Fixed	0.002073	0.002078
Time	0.5	Width only	0.000015	0.000015
Space	0.5	Width only	0.002735	0.002730

All new absolute-spread comparisons meet 0.01: True. Maximum spread-response differences are 0.00001486 (time) and 0.00439334 (space). These finite comparisons do not establish a convergence order or the continuum limit. No new response tolerance is chosen. Both phases are retained. Maxima use common 0.02 output times and post-event states at children, not every update.

Each programme fills 0.3 with unfilled volume below 10^{-12} . Maximum field/ledger errors are $3.6 \times 10^{-15}/6.2 \times 10^{-15}$. Accounting and admissibility are checked every update. The assessment also checks matched pending stocks and placement formulas. Fixed-control effects are retained; centre-only and nonlinear interaction stay at their prior resolution.

No long ensemble is recomputed. Earlier full-tape spatial refinements changed midpoint/spread ACFs by 0.10340/0.12893; fine lag-one midpoint ACF was 0.636–0.651. Whitening and limiting impact laws remain unestablished. Review measured response accuracy before registering another extension. Numerical/scientific acceptance remains open.

Paper and lineage. Retain the supplied long paper v1.1.9 and current SpreadLetter v1.2.0. The letter authors are Christopher Angstmann, Derick Diana and Tim Gebbie. Reaction price differs from quoted midpoint; distributed outward placement is not replaced by its narrow limit. Density response, first passage and completion lag remain distinct. No recurrence or manuscript is altered. The v0.5.5 replenishment wording correction is retained: buys initiate bid replenishment and sells initiate ask replenishment.

7 Future memory and clock compatibility

Keep four mechanisms distinct: event-time order splitting (β), operational transport memory (α_u), round-trip completion (h, τ_h) and calendar observation/subordination (α_t). Give stochastic components separate recorded seed streams. The event/operational/calendar distinction also follows the companion account [6]; its abstract provides context here, not verification of a fractional two-side solver.

The prior raw Sibuya kernel is

$$K_1 = \alpha_u, \quad K_2 = \frac{1}{2}\alpha_u(\alpha_u - 1), \quad K_k = K_{k-1} \left(1 - \frac{2 - \alpha_u}{k}\right), \quad k \geq 3.$$

The prototype applies survival $e^{-\nu(k-1)\Delta u}$ to lagged transport and $e^{-\nu\Delta u}$ to the current survivor, once each. Cancellation is not already folded into the raw kernel. $\alpha_u = 1$ leaves only $K_1 = 1$. Its scaling is $D_{\alpha_u} = r\Delta x^2/(2\Delta u^{\alpha_u})$; changing α_u while silently retaining an incompatible $r, \Delta u, D$ is invalid.

The current state has no transport-history buffer. A future fractional two-side model must derive the survival and history treatment of nonlinear reaction, cancellation, births and consumed volume. It must not resurrect removed liquidity through old history. This requires a stated formulation and tests against the prior primitive and its Markov limit; inserting the kernel into the existing Laplacian alone is insufficient.

The paper's stationary placement construction requires a finite mean completion lag. A heavy-tailed transport waiting time is not permission to replace the completion density with an infinite-mean law. That would change the equilibrium assumptions and can make pending exposure diverge under sustained input.

The correlation-emergence observation layer acts on completed operational paths. A future projection must select the previous completed state under the declared clock, retain index maps and reject out-of-support queries. Do not interpolate across an execution. Use a coherent book observation for bid, ask and midpoint before testing separate reporting delays. A held last trade price creates neither another trade nor another sign. Empty calendar bins have zero signed flow, not a repeated aggressor; distinguish them from missing pre-first-trade observations. Compare shocked/control paths using the same realized clock. These rules are specified, not yet implemented.

References

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